AWG19 POLARIZATION DRIVER — CONNECTED SCHEMATIC

NUCLEO-F446RE: see NUCLEO_WIRING.md for cable pin numbers.

U3 LTC1693-5 — R2 22 — R5 100k — Q4 IRF9640 — +12V — DR4 — TOP

U5 SN74AHCT125PWR — POL, TG, RX, SH, DP, RESET, POL, TG

R11–R14 100k — R15–R18 10k — C9 100n / 16V

J3 NUCLEO_F446RE_CABLE — POL_3V3, TG_3V3, RX, SH, DP, RESET, I_ADC, FF, +3V3, +5V

J2 TO COIL — COIL_A, COIL_B

G6K-2P DC5 — +5V, COIL_TOP, SIG+, BOTTOM, SIG−, RELAY_LOW, COIL_A, COIL_B

J4 TO INAMP — SIG+, SIG−, GND

J7 DRIVER_TEST_ONLY — TOP, BOTTOM

R9 0.25 / 2W / 1% — Q5 IRF530 — COIL_TOP — R6 100k — GND

SMBJ28A D1 — CLAMP_MID — US1D-13/H2

Q4 / Q5 need MCU dead time.
No hardware interlock yet.
Regulated 12 V supply required.

U5: 3.3V to 5V for gate drivers.
R11–R16: GPIO low when undriven. R17/R18: driver input pulldowns.
Power 3.3V and 5V from Nucleo only. J1–3 is a monitor/output.
No hardware interlock: firmware must prevent POL and TG overlap.

U1 LTC1693-1 — R1 22 — DR1 — DR5 — R4 100k — R3 2.2 — Q1 IRF530 — GND

BOTTOM — U2A ADG5412F — C1 1u — RX_IN — J6 LEGACY_RX_TEST_ONLY

R7 10k — DAMP — U2C ADG5412F

U2B ADG5412F — SH

U2D ADG5412F — RESET

U2E ADG5412F — R8 10k — FF

R20 1k — Q6 2N3904 — RELAY_LOW — D3 1N4148 — R21 10k — C10 100u 16V

J1 12V_INPUT_/_3V3_MONITOR — +12V_REG, −12V, 3V3_OUT, GND

U4 INA241A1IDR — TOP, COIL_TOP, +3V3, GND — +5V — CS_OUT — R10 1k / 0.1% — I_ADC — C2 10n — R19 1.5k / 0.1%

J5 RAW_SHUNT_SENSE — CURRENT_SENSE+ TOP, CURRENT_SENSE− COIL_TOP

J5: 250 mV/A differential. Neither pin is ground.
Use J3-7 / PA0 for MCU; J3-8 / PC7 = /FF.

INA241A1: gain 10, +5V supply.
I_ADC = 0.99 V + 1.5 V/A nominal.
Calibrate zero: current-low then validdated settling delay.

REV I: full coil-current feedback: not receiver-ready detection.
R9 is shared by polarization and clamp discharge.
J5 is HIGH COMMON MODE: never connect directly to MCU.
Use amplified I_ADC at J3-7.

LOCAL BYPASS: place each capacitor at its named IC supply pin

C3 1u U1-8 — C4 1u U1-6 — C5 1u U3 — C6 100n U2 — C7 100n U2 — C8 100n U4

REV I: ONE BOARD, FLOATING COIL RECEIVE
J2 is the coil. J4: 1 SIG+, 2 SIG−, 3 GND.
J7/J6 are DNP driver-side test connectors, NOT coil/receiver outputs.
RX=0: relay connects coil to polarization/discharge circuit.
RX=1: relay transfers both coil ends to SIG+/SIG−.
POL must be OFF and coil discharged before RX changes.
Wait for relay bounce and receiver settling in both directions.
INA818 requires input bias return and protection on receiver board.

ECE 455 — Jun Yang
Sheet: /
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Title: AWG19 Polarization — Connected Circuit
Size: A2 | Date: | Rev: I — 100x100 INTEGRATED
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